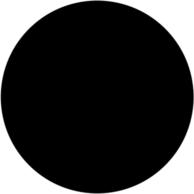
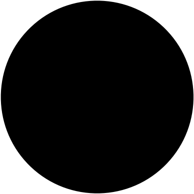
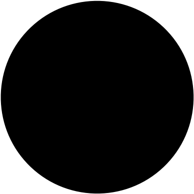
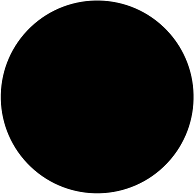
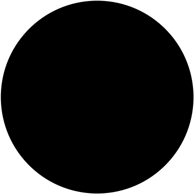










$$J = USV^T$$

$$\tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$\tilde{U}^T J \tilde{V}$$

f

11 x_1^{l-1} 1

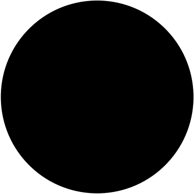
1

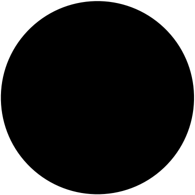
f

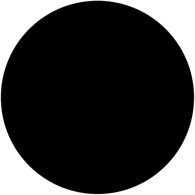
11 x_1^{l} 1

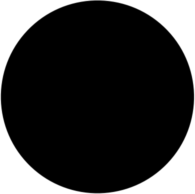
1

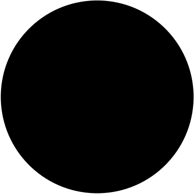
2

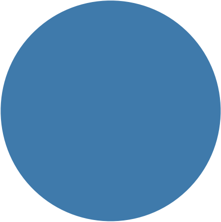


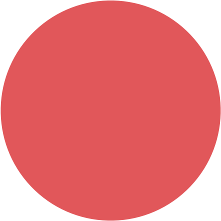


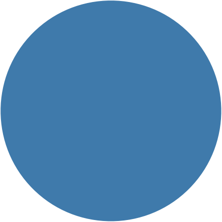


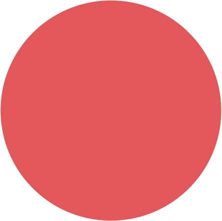


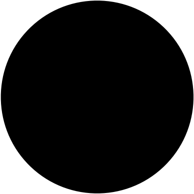












f

11 x_1^{l-1} 1

12

f

22 x_2^l 2

Constructing J

$$J(X) = \frac{\partial f}{\partial X}(X)$$

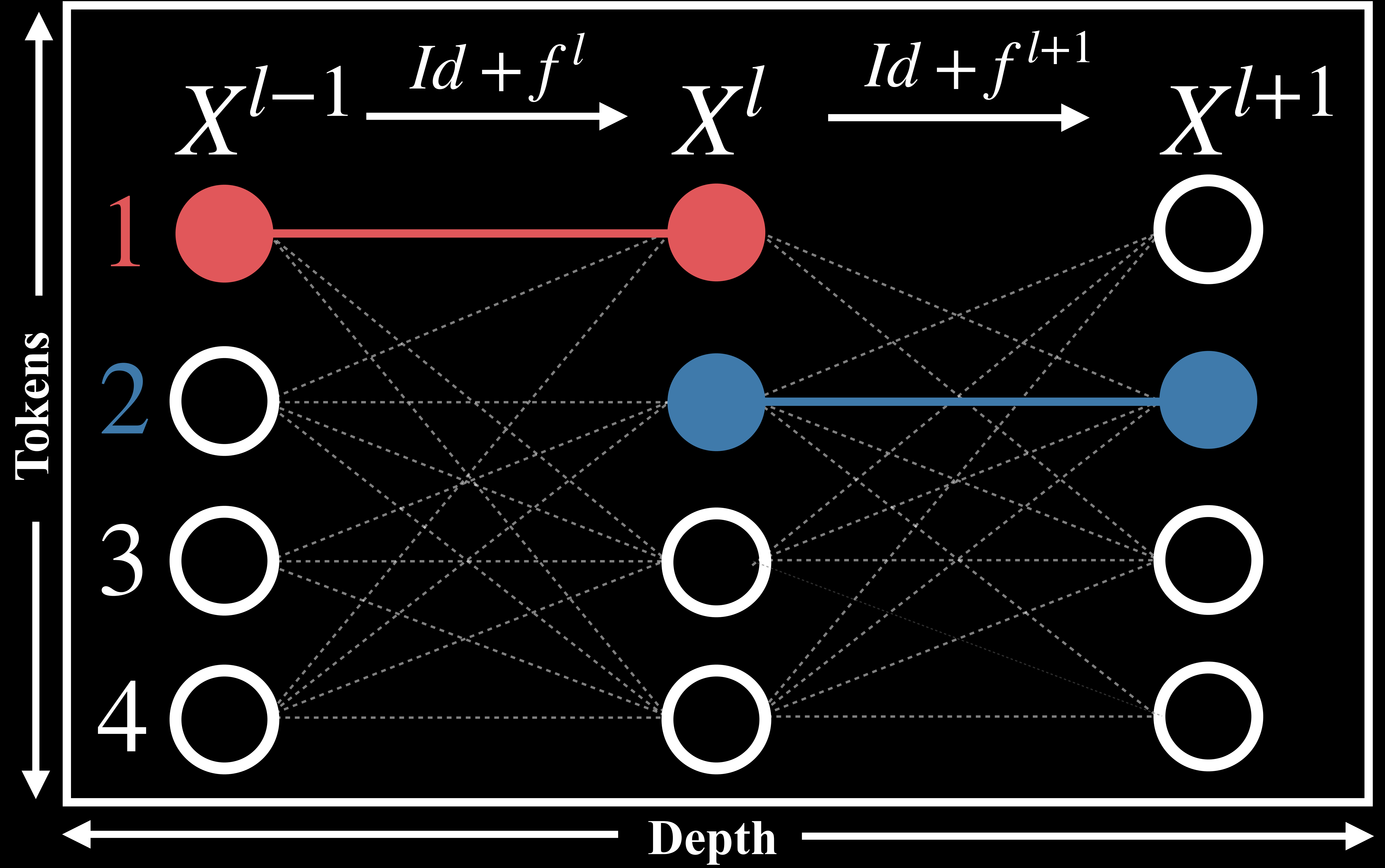
Constructing A

$$J = USV^T \quad \tilde{J} = \tilde{U}\tilde{S}\tilde{V}^T$$

$$A = \tilde{U}^T J \tilde{V}$$

Depth-wise Coupling

$$\left.\begin{aligned} J_{11}^{l} &= \frac{\partial}{\partial x_1^{l-1}}\left(f^{l}(X^{l-1})\right)_1 \\ J_{11}^{l+1} &= \frac{\partial}{\partial x_1^{l}}\left(f^{l+1}(X^{l})\right)_1 \end{aligned}\right\} A_1^{l,l+1}$$



Token-wise Coupling

Self coupling

$$\left.\begin{aligned} J_{11}^{l} &= \frac{\partial}{\partial x_1^{l-1}}\left(f^{l}(X^{l-1})\right)_1 \\ J_{22}^{l+1} &= \frac{\partial}{\partial x_2^{l}}\left(f^{l+1}(X^{l})\right)_2 \end{aligned}\right\} A_1^{l}$$

Token-wise Coupling

Context coupling

$$\left.\begin{aligned} J^{l}_{12} &= \frac{\partial}{\partial x_1^{l-1}}\left(f^{l}(X^{l-1})\right)_2 \\ J^{l+1}_{34} &= \frac{\partial}{\partial x_3^{l}}\left(f^{l+1}(X^{l})\right)_4 \end{aligned}\right\} A^{l,l+1}_{1234}$$